

```

|----- MODULE appex4_1_1 -----|
| EXTENDS Integers, TLC |
|-----|
| CONSTANTS x0, y0, z0 |
|-----|
|  $pre(a, b, c) \triangleq a = 10 \wedge c = 2 * a \wedge b = c + a$  |
|-----|
| Verifying the existence of a triple that satisfies  $pre(x0; y0, z0)$  |
| ASSUME  $pre(x0, y0, z0)$  |
|-----|
| --algorithm test {
| variables  $x = x0, z = z0, y = y0, t;$ 
| {
| l1: assert  $x = 10 \wedge z = 2 * x \wedge y = z + x;$ 
|  $y := z + x;$ 
| l2: assert  $x = 10 \wedge y = x + 2 * 10$ 
|   skip;
| }
| }
| BEGIN TRANSLATION ( $chksum(pcal) = "cec661f" \wedge chksum(tla) = "51e99e7b"$ )
| CONSTANT defaultInitValue
| VARIABLES  $x, z, y, t, pc$ 
|
|  $vars \triangleq \langle x, z, y, t, pc \rangle$ 
|
| Init  $\triangleq$  Global variables
|            $\wedge x = x0$ 
|            $\wedge z = z0$ 
|            $\wedge y = y0$ 
|            $\wedge t = defaultInitValue$ 
|            $\wedge pc = "l1"$ 
|
| l1  $\triangleq$   $\wedge pc = "l1"$ 
|            $\wedge y' = z + x$ 
|            $\wedge pc' = "l2"$ 
|            $\wedge \text{UNCHANGED } \langle x, z, t \rangle$ 
|
| l2  $\triangleq$   $\wedge pc = "l2"$ 
|            $\wedge \text{TRUE}$ 
|            $\wedge pc' = "Done"$ 
|            $\wedge \text{UNCHANGED } \langle x, z, y, t \rangle$ 
|
| Allow infinite stuttering to prevent deadlock on termination.
| Terminating  $\triangleq pc = "Done" \wedge \text{UNCHANGED } vars$ 
|
| Next  $\triangleq l1 \vee l2$ 
|            $\vee Terminating$ 

```

$Spec \triangleq Init \wedge \Box[Next]_{vars}$

$Termination \triangleq \Diamond(pc = \text{"Done"})$

END TRANSLATION

ASSUME pre

preliminaries for checking on possible values

$MAX \triangleq 32767 \quad 16 \text{ bits}$

$D \triangleq -32768 \dots 32767$

$DD(X) \triangleq (X \neq defaultInitValue) \Rightarrow (X \in D)$

$Inv \triangleq$

$\wedge pc \in \{\text{"I1"}, \text{"I2"}, \text{"Done"}\}$

$\wedge x \in Int \wedge y \in Int \wedge z \in Int$

$\wedge pc = \text{"I1"} \Rightarrow x = 10 \wedge z = 2 * x \wedge y = z + x$

$\wedge pc = \text{"I2"} \Rightarrow x = 10 \wedge y = x + 2 * 10$

$\wedge pc = \text{"Done"} \Rightarrow x = 10 \wedge y = x + 2 * 10$

$Safety_Partial_Correctness \triangleq pc = \text{"Done"} \Rightarrow x = 10 \wedge y = x + 2 * 10$

$Safety_rte \triangleq DD(x) \wedge DD(y) \wedge DD(z) \wedge DD(t)$

$check \triangleq Inv \wedge Safety_Partial_Correctness \wedge Safety_rte$

$prop \triangleq \Box(x = x0)$
